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WEIGHBRIDGE SYSTEMS & INSTRUMENTATION
Weigh Forward

CONVEYOR BELT SCALE ASSET REGISTER & FRAME-TYPE ASSESSMENT

Belt Scale Inventory & Frame Comparison

REPORT NUMBER ISS-BS-2026-001	**DATE OF ISSUE** 12 August 2026
CLIENT Woestaleen Colliery	**SITE TYPE** Colliery — plant conveyor scales
SCOPE 7 conveyor belt scales	**PREPARED BY** JP Pretorius — Founder & Technical Director

1 Overview

Woestaleen Colliery operates **seven conveyor belt scales** across the ROM and plant circuits. Although the seven installations differ in belt width, load-cell rating and speed sensor, they are built on just **two physical frame types** and read by **two indicator types**.

- **Frame Type A — Single-idler frame** weighing on **two single-ended “Surebeam” shear-beam load cells** (one each side).
- **Frame Type B — “Rhino” frame**, fixed at one end on threaded suspension rods and measured at the opposite edge by a **single centre S-type load cell**. It carries two idlers but, as noted below, is not a true two-idler weighment.

The two indicator types on site are the **Supmeter** (ROM scale only) and the **Loadtech LT1320** (the remaining six). The full register is given in Table 1, followed by an illustration and pros/cons of each frame.

2 Belt Scale Register

Table 1 — Woestaleen Colliery conveyor belt scales

#	Scale	Belt	Frame type	Indicator	Load cell(s)	Speed sensor
1	ROM scale	1050 mm	Rhino (2-idler*)	Supmeter	1 × 300 kg S-type	Rhino
2	Phase 1 feed to plant	900 mm	Single-idler	Loadtech LT1320	2 × 200 kg Surebeam	Loadtech
3	Phase 2 feed to plant	900 mm	Rhino (2-idler*)	Loadtech LT1320	1 × 300 kg S-type	Rhino
4	Phase 1 product	750 mm	Rhino (2-idler*)	Loadtech LT1320	1 × 300 kg S-type	ISS
5	Phase 2 product	750 mm	Single-idler	Loadtech LT1320	2 × 200 kg Surebeam	Loadtech
6	Phase 1 discard	750 mm	Single-idler	Loadtech LT1320	2 × 200 kg Surebeam	Loadtech
7	Phase 2 discard	750 mm	Rhino (2-idler*)	Loadtech LT1320	1 × 300 kg S-type	Rhino

* Carries two idlers on the weigh frame but measures through a single pivoted S-type cell — not a true two-idler weighment (see Section 4). Split: **3 × single-idler** (Phase 1 feed, Phase 2 product, Phase 1 discard) and **4 × Rhino** (ROM, Phase 2 feed, Phase 1 product, Phase 2 discard).

3 Frame Type A — Single-Idler (Two Surebeam Cells)

A single weigh idler is carried on a short frame that rests on **two single-ended shear-beam (Surebeam) load cells**, one on each side. The full weight of belt and material on that idler passes straight down through the two cells, which are summed by the indicator — a direct, symmetrical weighment.

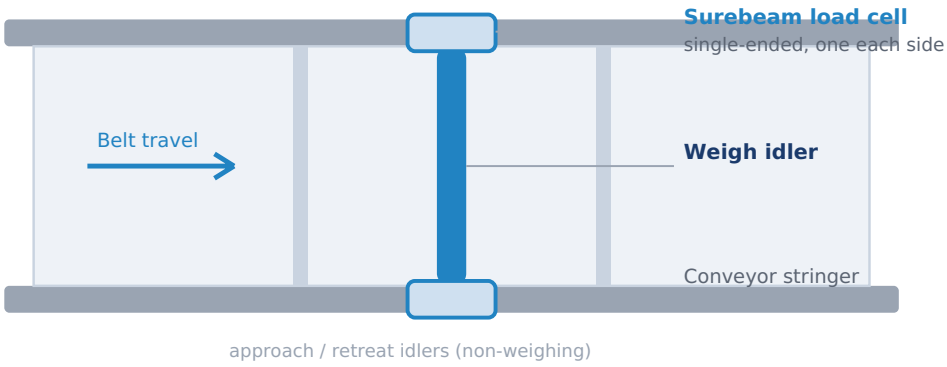


Figure 1. Single-idler frame — plan (top) view. The weigh idler spans the belt and is carried at each side on a single-ended shear-beam (Surebeam) load cell mounted to the stringer; the two cells are summed for a direct weighment.

4 Frame Type B — “Rhino” Suspended Frame (Single S-Type)

The Rhino frame carries **two idlers**, but it is not weighed like a true two-idler scale. One edge is **fixed and suspended on a threaded rod at each of its two corners**; the opposite edge hangs at its **centre on a single S-type load cell** (in tension). The rod end acts as a pivot, so the single cell reads only a **geometry-dependent share** of the load — the frame behaves closer to a pivoted single-point scale than a true two-idler weighment.

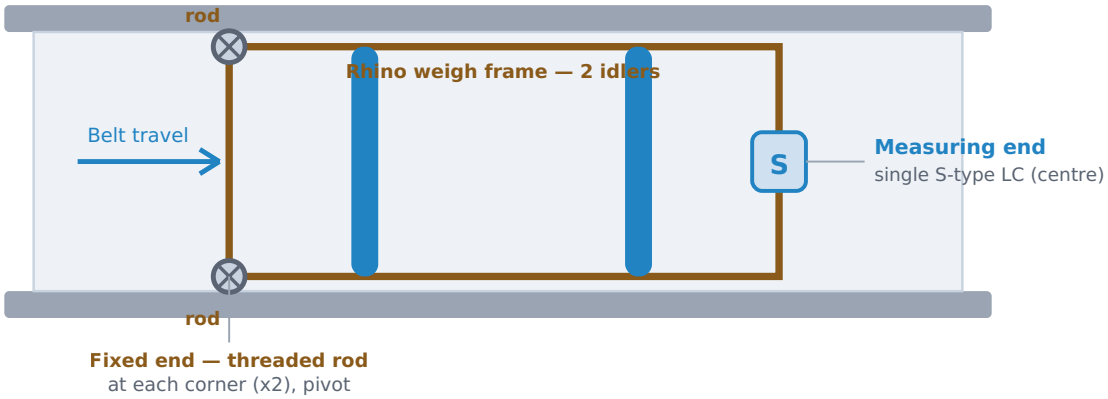


Figure 2. Rhino frame — plan (top) view. The frame carries two idlers across the belt; its fixed edge hangs on a threaded rod at each corner (a pivot) while the opposite edge is measured at its centre by a single S-type load cell — so that one cell reads only a geometry-dependent share of the load, not a true two-idler weighment.

5 Pros & Cons of Each Frame

Frame A — Single-idler

Two Surebeam shear-beam cells

PROS

- Full idler load passes straight down through two cells — a direct, true weighment.
- Two cells average side-to-side, so off-centre / mistracking belt load largely cancels out.
- Rugged, standard shear-beam cells; simple to install, calibrate and fault-find.

CONS

- Short weigh length (one idler) → more sensitive to belt tension, idler alignment and local belt effects; needs well-aligned approach/retreat idlers.
- Single-idler scales sit at a lower inherent accuracy class than a true multi-idler for the same duty.

Frame B — Rhino

Suspended frame, single S-type cell

PROS

- Carries two idlers, so a longer length of belt is supported on the weigh frame.
- Single S-type cell in tension is a clean, low-hysteresis measurement; the hanging frame is isolated from some structural stress.

CONS

- Not a true two-idler weighment — the pivoted rod end carries part of the load, so the single cell sees only a geometry-dependent share.
- One cell only → no side-to-side averaging, so belt mistracking skews the reading.
- Threaded-rod suspension can bind, corrode or shift in length/tension, and the pivot must stay free — all of which move calibration and hurt repeatability.
- More sensitive to frame level and rod adjustment; harder to hold in calibration.

6 Observations & Notes

- **Frame split:** three single-idler Surebeam frames (Phase 1 feed, Phase 2 product, Phase 1 discard) and four Rhino frames (ROM, Phase 2 feed, Phase 1 product, Phase 2 discard).
- **Indicators:** two types — Supmeter on the ROM scale only, and Loadtech LT1320 on the other six.
- **Speed sensors are mixed** across the site: Rhino (ROM, Phase 2 feed, Phase 2 discard), Loadtech (Phase 1 feed, Phase 2 product, Phase 1 discard) and ISS (Phase 1 product).
- **Metrological note:** the single-idler Surebeam frames are the sounder design — a direct two-cell weighment. The Rhino frames, though they carry two idlers, measure through one pivoted S-type cell and should be expected to be more sensitive to mechanical condition (rod/pivot freedom, frame level) and lower in inherent accuracy and repeatability.

Prepared and inspected by:

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This register reflects the belt scales and configuration observed on site at the time of inspection. Frame illustrations are schematic and not to scale. Load-cell ratings, indicator and speed-sensor types are as recorded per scale in Table 1. Accuracy commentary is a general engineering comparison of the two frame designs and does not replace a per-scale calibration and verification.